

ALLPLAN 2022

New Features in Allplan 2022

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Allplan 2022 – Buildability at its best

Optimal Buildability through seamlessly integrated processes

Allplan 2022 stands for the seamless working method of architects, engineers and contractors on a single platform from the initial design to successful implementation on the construction site. All information such as floor plans, elevations, sections, reports, quantities or costs are available to all project partners. Quality losses due to imports and exports during system changes are a thing of the past. This shortens coordination processes, increases efficiency and enhances added value. **Allplan 2022** provides you with a seamlessly integrated process in line with the BIM way of working – for buildability at its best.

Right at the start of the project, **Allplan** supports thorough capture of the construction environment with functionality for the terrain model and road design. In the course of the project, you can increase your efficiency and precision with the help of optimized modeling tools. Visualizations become even more realistic and of higher quality. The areas of concrete and steel structures have been further developed. **Allplan 2022** offers new functions for implementation on the construction site. At the same time, the new version impresses with optimized user-friendliness in many areas.

Optimizations for Installation and Project Import

You can now get started even faster with the new Windows Installer. New design and a simpler user interface: Faster download and installation process.

If you have a large number of workstations, you can save all settings and then run the installation completely automatically. In addition, you can now import projects from any source directly in the project selection dialog. You can activate or deactivate the **Allplan Workgroup Manager** at any time.

Allplan Installation Significantly Simplified

Allplan installation simplified considerably. As a user, the settings you make during the installation are now limited to a few specifications, such as defining the program path and the data path, and selecting the options to be installed. For an upgrade installation, you only have to choose between upgrading an existing Allplan installation and a new or parallel installation of Allplan 2022.

You can make all other settings and transfer settings, projects and other data from previous versions downstream using **Allplan Diagnostics** (select in **Allplan**:  **Help** menu; select in **Services application**: **Service** menu) or direct project import via  **new project, open project**.

Important information about the Workgroup Manager!

When you reinstall Allplan 2022, you no longer need to explicitly specify whether to install with or without the Workgroup Manager. The Workgroup Manager is installed *automatically* as of version 2022.

If you want to run Allplan in a network *with* Workgroup Manager and you have a corresponding license, activate the Workgroup Manager via **Allplan Diagnostics** (**settings** tab, **workgroup settings** area). For more information, see "Allplan Diagnostics" in the Allplan Help.

Import projects, also from previous versions

In the newly designed dialog box **new project, open** you can now also import projects or zipped project backups. This way you can easily access projects from all supported Allplan versions.



importing projects

Here you can import one or more projects or zipped project backups from any path. The project can also originate from an earlier, still supported Allplan version. You can select the projects in the **import project** dialog box and import them into the current version. The source version is displayed in the **version** column; the project is converted to the current version when you first open it. You can use this function to upconvert projects from previous versions after upgrading to the current Allplan version.

Tip: You can find the path where the local projects are stored via **Service application – service – Windows Explorer – my own CAD projects (PRJ)**.

Allplan 2022: easy operation right from the start

To make it easier to get started with **Allplan**, the tooltips have been enhanced: when you hover over an icon with the mouse button, a brief explanation of the function is now automatically displayed. Developments to standardize the user interface have also been continued: For example, property palettes have been added for more object types such as openings and steel connections components. For the simplified use of cloud services, there is now a common login for **Allplan Connect** and **Bimplus**. Another new feature is the ability to rotate around a selected object in the animation.

The demo project "Hello Allplan!"

HELLO ALLPLAN! – Your quick introduction to how to get started with Allplan.

The demo project "Hello Allplan!" is automatically installed with Allplan. It is intended to make it easier for you to get started with Allplan using an example project by allowing you to conveniently try out using the tools on existing objects. Nothing can go wrong, because you can delete and reinstall the demo project at any time.

The demo project is supplemented by the **Hello Allplan! videos**, which are adapted to the project and take you through the diverse possibilities of the model-based planning method. You can access the videos via the **HELLO ALLPLAN!** (<https://www.allplan.com/hello-allplan/>) website.

You can find the demo project "Hello Allplan!" after starting Allplan in the ALLPLAN welcome dialog. Just select and start!

Properties palette

To customize your model even more effectively and quickly, you can now modify complex window and door openings using the **properties** palette. You use uniform workflows and, in addition to geometric properties, you can also change reveal, parapet, stop or sill types as well as integrated objects such as Shading SmartParts here. In addition, you can modify all relevant window or door openings in one step via the **properties** palette by activating the desired openings and entering the changes in the **properties** palette. As a result, all activated openings receive the changes made in one step.

Modify openings

When you activate one or more door or window openings, the **properties** palette now offers all the parameters that are also contained in the **door** or **window** dialog boxes. This allows you to quickly and easily change one or more openings in one step.

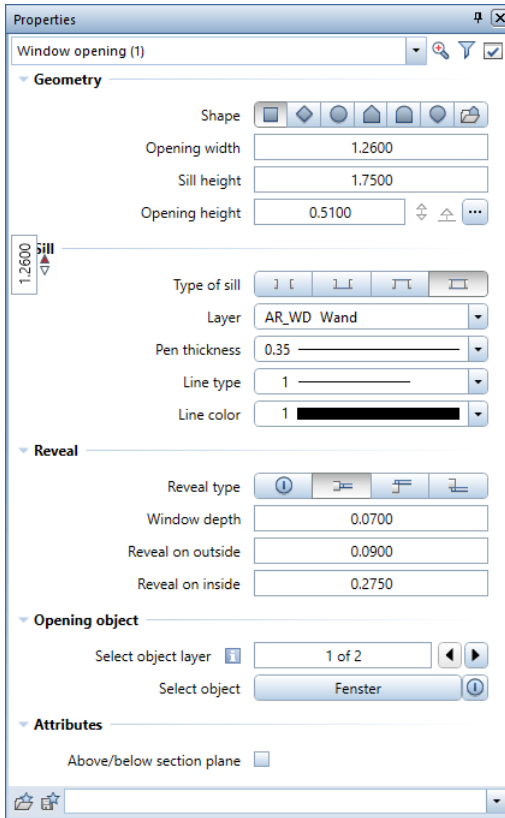


Figure: Palette properties for modification, one window activated

Overlapping Openings

The **above/below cutting plane** option, initially offered only in the **properties** palette, lets you specify how overlapping openings (windows, doors, niches) are displayed.

If the option is activated, the opening in the wall will not be cut out, but only displayed as an outline; the wall hatching will be preserved. You can assign the format properties of the opening and opening object individually. This creates a correct representation of overlapping openings in the floor plan and in the layout.

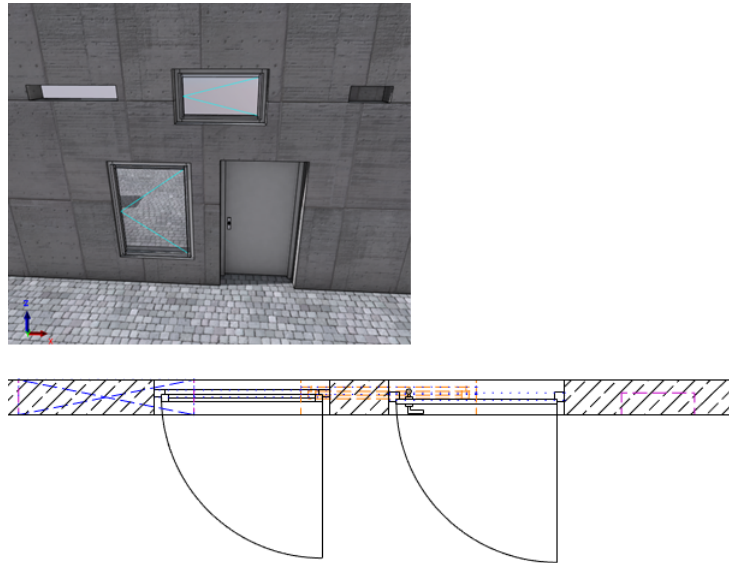


Figure: Display openings above the cutting plane

Enhanced Properties Palette

The palettes for some functions have been integrated into the **properties** palette. For the relevant functions, the *additional* palette that was previously displayed for defining the settings is now displayed *instead* of the **properties** palette.

In detail, these are the palettes for the following functions:

- **view type** box (viewport toolbar)
-  **3D view** box (viewport toolbar)
-  **surroundings** (surfaces, **light** task area)
-  **assign custom surfaces to 3D, archit. elements** (surfaces, **light** task area)
-  **set surface** (surfaces, **light** task area)
-  **render** (surfaces, **light** task area)
-  **set project light** (surfaces, **light** task area)
-  **luminaire** (surfaces, **light** task area)
-  **set luminaire** (surfaces, **light** task area)
-  **set camera path** (camera task area)
-  **record film** (camera task area)

Layers Palette

You can now sort layers alphabetically in the **layers** palette. To do this, click the **short name** or **long name** column labels in the list header. There are three positions for sorting. After the first click the sorting is done from A to Z, after the second click from Z to A and at the third click the default sorting is restored.

Activate a single wall layer

Two clicks used to be required to activate a single layer of a multilayer component: With the first click you could select the whole component, with a second click you could select the desired layer. If there was not enough time between the two clicks, the system interpreted this as a double click and opened the **properties** dialog box.

New: You can select individual wall layers with a single click by holding down the SPACEBAR and clicking directly in the desired wall layer. The wall layer activated this way is immediately selected in the **properties** palette and can be edited.

Roof Surface

When creating a  **roof area**, all relevant parameters are now immediately displayed in the **properties** palette.

In the **properties** palette, in the **edge settings** area, the **edge selection** has been added. When creating the roof area, you will be informed here which roof edge is being created.

After completing the contour, you can select the desired edge or several edges (e.g. 1-3 or 1,4) in the input field by entering the edge number. Use the arrow keys to select the previous edge of the currently displayed edge or the edge following it.

Options

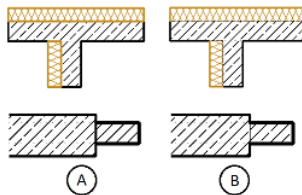
New option 'fixed pen and color for junction, division lines'.

Under  **options – components and architecture – components** you will now find the new **fixed pen and color option for junction and division lines**. This allows you to define how junction and division lines between components are displayed.

- If the option is deactivated (default), then the format properties of junction and division lines are taken from the respective part.
- If the option is enabled, you can set a fixed pen and color with which the junction and division lines will be displayed.

You can use this option to adapt the display to country-specific standards, for example.

Note: This option is a general **Allplan** setting and applies globally to all projects.



- A: Option deactivated; pen/color of junction and division lines like component
B: Option activated; junction and division lines with fixed pen/color

New option 'Automatically Update Views and Sections after Layout Editor'.

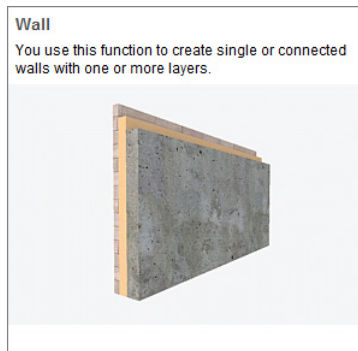
To avoid having to decide whether to update views and sections referenced to the model data every time you switch to the layout editor when the model data is changed, you will find the new **layout editor** section, which replaces the dialog box display, under  **options – views**.

Basically, all views and sections are always updated when switching to the layout editor. If model data has been changed without activating the linked views and sections, the representation in the views and sections may not be correct. If the **automatically update views and sections by layout editor** option is enabled, the representation will also be updated in such views and sections. When switching to the layout editor, you will be informed that you can adjust the automatic update.

If, on the other hand, the **automatically update views and sections by layout editor** option is deactivated, layout elements containing views and sections in which the model data are not correctly displayed will be displayed with the background color set under **background color of obsolete views**. You can manually update such layout elements using the  **update layout** function.

Extended ToolTips

Extended ToolTips consist of a short description of the function; they can also contain further information, illustrations and/or links to short explanatory videos. You can use them for support when getting started with **Allplan**. To show them, keep pointing the cursor at an icon if the corresponding ToolTip is already shown.



Extended ToolTips are enabled by default; They are activated or deactivated in the same way as the ToolTips with **customize user interface – customize** tab (**quick access toolbar**).

Note: Extended ToolTips are only displayed if **actionbar configuration** is enabled.

Copy drawing file or layout to several drawing files or layouts

Now you can copy a drawing file to several drawing files in one step in the building structure; this can help when you create variants. In the building structure, right-click a drawing file and then click **copy** on the shortcut menu. Then activate several empty drawing files (e.g. by holding down the CTRL key), right-click in the selection and then click **paste** in the context menu: The drawing file is copied from the clipboard to all activated target drawing files; the drawing file name is retained. Copying the drawing file to existing drawing files is not possible.

For layouts elements: In the layout structure, right-click a layout and then click **copy** on the shortcut menu. The layout must not be active. Then activate several layouts e.g. in the structure level **all layouts** (e.g. with pressed CTRL key), right-click in the selection and then click **paste** in the context menu: The layout is copied from the clipboard to all activated layouts. In contrast to the drawing file, the activated layouts can also already be occupied.

If there are already occupied layouts in the activated layouts, you can decide whether these should also be replaced. You can copy the layout name as an option in this case. If you have activated only empty layouts, the layout content including the layout name will be inserted in all layouts without any further query. Existing layout names will be replaced by this.

Note: This option is only available in the building structure or the layout structure, but not in **ProjectPilot**.

Reorganization of Layer Structure

The layer structures supplied with Allplan have been extended with the following new layer structures: ALLPLAN GENERAL and INFRASTRUCTURE.

ALLPLAN GENERAL contains the layer groups **surface elements, text, dimension line, views and sections, layout, visualization** and the general layers of the **construction** layer group. So far, you could find these layergroups under the layer structure ARCHITECTURE.

The layer structure INFRASTRUCTURE with the layer groups **general, temporary, measures, lining, components, covering and equipment** has been added.

Smart Catalog

The Smart Catalog offers you the possibility to create your own material catalog. You can select this XML catalog when you create your project, but you can also assign it afterwards in the project properties (path settings – CAD-TAI project assignment). The material data accumulated in the catalog is then available to you in the material selection for your objects.

The Smart Catalog contains some sample data. You are not bound to the structure of this sample data in the setup of your catalog. You are free to choose the hierarchy as well as the type and number of your material data. If you have added data to your Smart Catalog, it will be stored in the office default folder ...\\STD\\XML Catalog Data after saving.

The call to edit the catalog is currently made via Service application – **service – smart catalog**.

Layout frame and title block for multiple layouts

Using the **assign title block** entry in the context menu of the plan structure, you can now assign a plan frame and/or a plan legend to all selected layouts in one step. If a layout frame is selected, you can also use a label style as title block. You can specify **offset to the right** and **offset to the bottom** for the layout legend and the title block. If you do not activate the **distance** option and no layout legend or title block for the selected layouts, the default values of 5.0 mm apply. If not, the existing distances will be maintained.

If you have already defined a layout frame and/or a layout legend or a title block for the selected layouts, these will be replaced by the assignments defined here.

Improved 3D navigation

You can now move more easily and intuitively in navigation mode, e.g. in the animation windows. This is thanks to the following new features:

Rotation around an element

When rotating in **sphere mode** (with the left mouse button pressed), the camera was previously rotated around the axis of the scene or model. Now you can set the position of the rotation axis as follows:

- If you click outside the model, the camera is rotated around the axis of the scene as before, i.e. around the entire model.
- When you click in the model, a coordinate cross is displayed at the clicked point and the camera is rotated around this point. The Z axis of the superimposed coordinate cross corresponds to the rotation axis of the scene. This makes it easier to move around indoors in particular: For example, you can click on a table in the middle of the room and then rotate the camera around it.

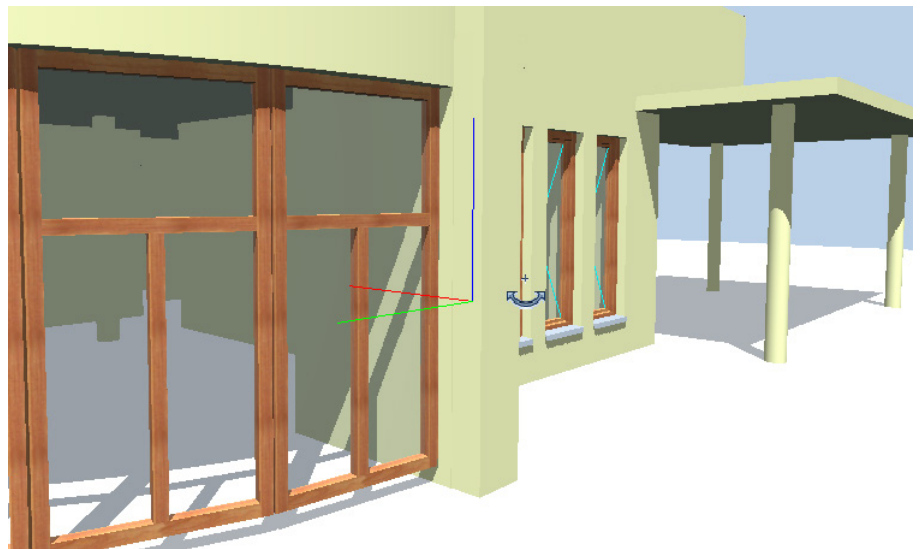


Figure: Rotation around a point of the model, with coordinate cross at the clicked point

Option – reverse direction in navigation mode

The option **reverse direction in navigation mode** is activated by default mode. The movement is now analogous to other programs.

You can find the **reverse direction in navigation mode** option in  **options – general – mouse and crosshairs** in the **mouse** section.

Other New Features

Start Screen

When you launch Allplan for the first time, the workspace opens in the  **2 + 1 animation window** arrangement. The graphic windows for view and animation, which are arranged one above the other, are positioned on the right. The **coordinate system** is displayed in the graphic windows.

Default configuration for window arrangement now 2 + 1 animation

The default configuration for the window arrangement has been changed to **2 + 1 animation windows**.

Title Bar

The  **Allplan status page** function has been added to the title bar of the Allplan application window. Furthermore, the  **help** pull-down menu has been extended with the **support remote** function and the existing entries have been rearranged.

"Allplan Status Page" Function

You can use the  **Allplan status page** function to obtain information about the status of the running systems, e.g. if you are connected to the ALLPLAN Cloud. When you select the **Allplan status page** function, the website <https://status.allplan.com> opens. In the  **actionbar configurator** (right side of the **actionbar**) you can find the new function under the **more functions** section.

"Support Remote" Function

The **support remote** function has been added to the pull-down menu of the  **help** icon (title bar of the Allplan application window). By activating this function, you can start a remote access software (e.g. TeamViewer) directly from Allplan.

"Support Remote Download" Function

Under **Allplan on the web** (drop-down menu of the  **help** icon), the **Allplan remote maintenance** function has been renamed to **support remote download**. The mode of operation remains the same: You can download a remote maintenance program.

Status Bar

The status bar of the Allplan application window has been extended by one element: the **notification center**.

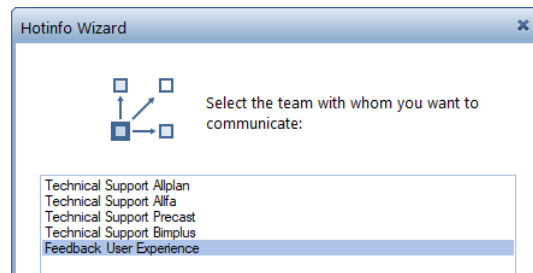
Under  **notifications**, you can see warnings, hints or errors concerning the model or the modeling. The aim of this is to help you improve and control the creation of your model. When there is a new important information, you will see the following icon  and the Notification Center message/notice will be displayed briefly at the bottom right corner of the screen. You can permanently show the message/note by clicking the changed icon in the status bar. If you do not want the messages/instructions to be displayed automatically for future actions, activate the **suppress message until project change** option. If errors listed in the Notification Center are corrected, the icon changes back to .

Note: Currently, the Notification Center only manages notices related to the positioning of structural framing objects. Other areas will be added in the future. Existing messages will also be displayed in the Notification Center in the future.

Feedback to User Experience Team

Do you have any comments about the user interface, operation or suggestions for improving **Allplan**? You can now contact our User Experience team directly via **hotinfo**, the Allplan support tool.

Start **hotinfo**, select **feedback user experience**, and follow the dialog with the program.

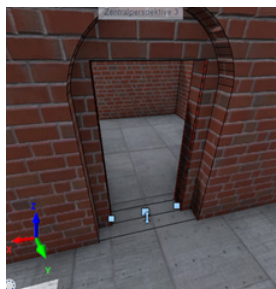


Label Polylines

The context menu of polylines created with  **offset polyline** (select) now offers the **label** function, analogous to the polylines created with  **offset 3D polyline** (select).

Niches

Complex openings in old buildings are often modeled by two adjacent niches. The common area of such niches is no longer filled with finishing surfaces of contiguous rooms.



Save Legends

When creating a legend, you will now receive a corresponding message if the ZLE file in which the legend is to be saved is too large. Alternatively, you can save the legend in another folder.

Replace layout depends on version

To replace a layout with another one from a folder or a ZIP file, you can now select layouts from the current version as well as from the last three previous versions.

Function for labeling the layout renamed

The function for labeling a layout with a label style has been renamed because misunderstandings repeatedly arose due to the name being the same as the  **label** function in the **annotations** task area. The function is now called  **label layout border**.

Empty Layout Windows will be Kept

If you set the layout window outside the associated layout elements when you create it or if you modify the layout window so that the associated layout elements are outside the layout window, you will now be left with an empty layout window that retains the information about the included layout elements; you can edit it at any time.

Faster and more secure collaboration with Allplan Share

Numerous optimizations have been made for work across locations with **Allplan Share**. For example, data is uploaded in the background without waiting times. In addition, there is a new backup function that can be used to automatically save revisions of drawing files and plans. If the Internet is not available or is disrupted, you can access the data in your computer's cache and simply continue working.

Autosave Drawing Files and Layouts

You can now specify how drawing files and layouts are to be saved automatically in the  **settings** of the dialog box **open on a project-specific basis: drawing files from fileset/building structure** or **open on a project-specific basis layouts**. You can disable saving by default and set the number of alterations.

The settings for drawing files and layouts are made separately and are independent of each other. They are saved in the respective project folder in the `AutoBackupSettings.xml` file. You can find the individual alterations of the drawing files and layouts in the **ndw** and **layout** subfolders in the **backup** folder of the respective project folder.

Note: To prevent duplication of data in projects, no alteration is generated during data conversion for drawing files and layouts.

More efficient and detailed modeling with Allplan

Detailed and precise models are a key prerequisite for efficient processing of BIM projects. With **Allplan 2022**, the interaction of components has been fundamentally revised. Intersection within a drawing file is now controlled consistently via priorities. Manual rework is no longer necessary. The resulting increased model quality reduces the effort required to create working drawings and detailed drawings. Collisions are avoided and quantities can be determined even more reliably.

In addition, the workflows for modifying openings and with roof surfaces have been simplified and optimized. The railing function has been revised so that railings can now be placed along splines. This opens up new areas of application, e.g. for organic architecture, but also for bridge railings or guard rails.

Intersection of Different Components via Priority

The values for Priority that you specify when creating a part not only affect objects of the same type within a drawing file (as before), but also control the intersection of different components with each other.

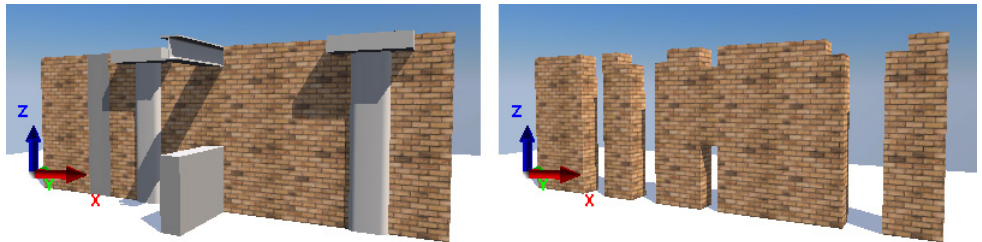


Figure: Intersection of different components in a drawing file via priority

- Intersection is performed regardless of the type of component and height (horizontal, vertical, different heights.)
- The intersection is based on the **Priority** property.
- The **priority** attribute can also be set in the **properties** palette or in the **modify attributes** dialog.
- You can also see the intersection in the floor plan (e.g. for different heights).

You can also change the **priority** afterwards in the **properties** palette: This has the advantage that the intersection of the component with all intersected elements is immediately recalculated without having to explicitly perform  **restore 3D view**.

Note: Until now, you could prevent the intersection of two components by using different relative heights at the bottom. This is now no longer possible; please use the **priority** instead.

Deactivate 2D interaction of Components

In the **properties** palette you will now find the new **2D interaction** option for components or individual layers of components. This allows you to control the interaction of architectural components with other components in the floor plan, the 3D representation is not changed.

The **2D interaction** option is active by default. If you deactivate the option, the components are no longer intersected in the 2D floor plan. Certain 2D lines can be hidden, for example.

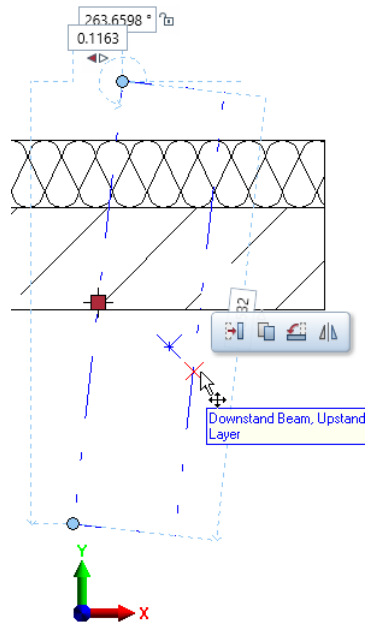


Figure: Option **2D interaction** of the beam is disabled

Writing your own scripts is becoming easier and easier

A number of optimizations in the **Visual Scripting** area make creating custom scripts even easier and faster. For example, the new palette designer allows parameters to be placed selectively on multiple tabs. Tooltips have been improved and sliders are now supported. It is also possible to access additional components such as columns and beams and to control IFC export. In addition, actions can be undone or repeated with just one click.

Visual Scripting

Visual Info

When you sort a parameter from the lower area to the upper area of the palette **palette designer**, you now get visual feedback about the drag and drop behavior.

If you drag the parameter onto the  **new group** icon or an existing tab or  for a new tab, these areas will be displayed with a colored background. Now place your parameter, it will be placed exactly at the selected location.

If you drag the parameter into the empty area after the last list entry of a tab, you will create a placeholder (**create new group**), in which you can then place your parameter.

When groups are  collapsed, the appearance of the expanders indicates whether the group contains elements (expander =  black filled triangle) or whether the group is empty (expander =  gray outlined triangle). You can also recognize a still empty group by the fact that the group name is displayed in italics and in the color gray.

If you have made any changes to the logic of your script, such as deleting or adding nodes, you will see a note about this in the properties palette of Allplan. You will be prompted to  **run** the script again in the visual scripting application window. As a result, you will see the adjusted parameters in the properties palette of Allplan.

Undo/ redo

The undo and redo functionalities are now also available in the visual scripting application window (toolbar and **edit** menu).

 **Undo** (CTRL+Z): Allows you to undo one or more actions.

 **Redo** (CTRL+Y): This allows you to restore actions that have been undone.

Modify series line

If you have accidentally disconnected a series line at the input port, restore the connection by clicking the ESC key.

The appearance of the series line now makes it easier to recognize its selection state:

- gray and narrow = normal and not selected
- light gray and narrow = mouseover
- dark gray and wide = selected and context menu is open

Duplicate node

When you **duplicate** (**edit** menu or CTRL+D) a single node, only its input connections are duplicated, the output connections are not.

Unlink Smart Symbol

The shortcut menu of a visual scripting object in Allplan now also offers the  **unlink smart symbol** function. You can use this tool to resolve the structure of the PythonPart relocation. You can then modify individually the design entities that make up the PythonPart installation.

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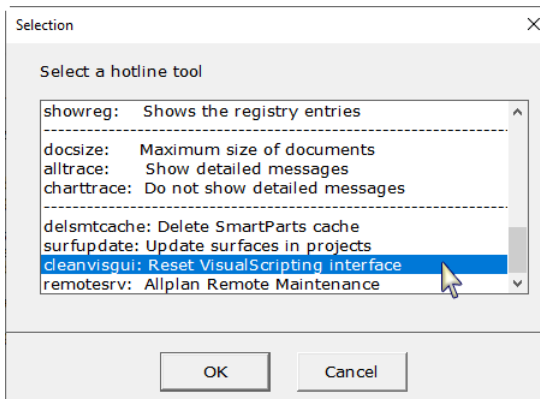
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New Hotline Tool cleanvisgui

Hotfix Allplan 2021-0-6 has extended the hotline tool **cleanup** (in **Services application – service – hotline tools**); it has also initialized the display of **visual scripting**. Allplan 2022 provides the **cleanvisgui** hotline tool for this purpose: Running it recreates the PypConWpfDlg.vspfile.xml and VisualEditor_WindowLayout.config files. This functionality is omitted in **cleanup**.



Automated Reinforcement: Fast and Precise

The automated reinforcement for beams, columns, walls and punching shear reinforcement introduced with **Allplan 2021-1** has been improved in many respects. As a result, the underlying PythonParts remain intact even after placement and can be parametrically adjusted at any time.

As a new feature, you can define user-specific bar shapes. The content of associative legends is precisely controlled via drawing file filters. A further upgraded feature is that you can copy reinforcement including sections and text labels. This saves a lot of time when displaying a large number of similar reinforcement situations.

Select Automatic Reinforcement Directly

You can now select the PythonParts for generating the automatic reinforcement for beams, columns and walls and the automatic punching shear reinforcement directly in the actionbar. In the **engineering structures** task, you will find the new **automatic reinforcement** task plane. Alternatively, PythonParts can still be opened in the **library** palette under **standard** ▶ **PythonParts** ▶ **automatic reinforcement**.

Shape Code Manager

Until now, the shape codes used in the legend **bar schedule – ACI** were hard-coded. You can now find these shape codes in the Etc directory in the **ShapeCodes** folder in the text file `ShapeCodes_ACI.txt`. This gives you the possibility to change the existing shape codes as well as add new shape codes.

If you want to use the changed shape codes only in the project or in the office standard, copy the text file into the **ShapeCodes** folder of the respective project folder or the Std directory. Please note that in the Etc directory the files are overwritten during an update.

When generating the legend, the search for the text file `ShapeCodes_ACI.txt` is performed in the **ShapeCodes** folder in the order project folder, Std directory and Etc directory. This means that the file can exist several times with possibly different contents, but it is only evaluated according to the defined priority.

Note: In the **ShapeCodes** folder of the Etc directory you will also find the text file `EndPreparations.txt`. In this file you will find all the entries that you can use to assign the new **end processing abbreviation** attribute. You can find this attribute in the **engineering** group. It is used in connection with the use of sleeves. The attribute was also added as a parameter in the article catalog when creating the sleeves and can be assigned accordingly. For this purpose, the file `Abbreviation_Endpreparation.lst` must be assigned in the **Lst** folder of the respective sleeve manufacturer.

Legends with drawing file filter

In the legends for the mesh, the reinforcing bar and the site plan, the elements of the selected drawing files are now taken into account even if these drawing files are not selected.

In the case of the general legend and the legend for thermal insulation, landscape planning and drawing symbols, on the other hand, the elements of the selected drawing files continue to be filtered only if the drawing files are loaded actively or passively.

Move Linked

Since moving views and sections together with the associated model data now also works with the normal  **move** edit function in such a way that the position of the elements in relation to each other remains unchanged, the  **move linked** function no longer applies.

Also copying, mirroring, rotating, ... of views and sections together with the associated model data now works with the normal edit functions.

Other New Features

Project Templates

You can now also use two project templates for civil engineering when creating projects. You have one variant in which the formwork and reinforcement are created together in one structural level and one in which the formwork and reinforcement are created on separate structural levels.

Labeling Options

Only administrators and project owners can permanently change settings in the reinforcement labeling options. Users without administrator rights, can only change the settings temporarily. This also applies to choosing the font.

Unit for Legend Break

If you enter a value in the dialog line for defining the break within the legend, it must now be entered in the set unit of length.

Breaks in Unicode Style

In order to get the most compact reinforcement labeling possible for imperial units, fractions are now used in Unicode style.

More variety in steel construction with Allplan

Based on the structural framing objects, bolted and welded steel joints have been newly developed for **Allplan 2022**. To accommodate the enormous variety, the structural steel joints are based on basic elements, e.g. for sheet metal, stems, bolts or welds. These are available in the new connection tools, but can also be combined into connections by Python scripts. The scripts are accessible to all users and can be adapted, extended or even completely redefined for other steel joints.

In addition, the axis grid on which many steel structures are based has been further improved in terms of representation and labeling. Standard or country-specific material catalogs can be downloaded via Bimplus and linked to components. In the case of structural framing objects, the brace type has been added. You can also position the objects automatically, even across different drawing files and when using the Allplan Workgroup Manager or Allplan Share.

Creating Steel Joints

With the new  **connection toolbox** function in the **structural framing objects** task area, you can open the PythonPart to create connections for structural framing objects directly in the actionbar. If you click on the function, the input palette of the PythonParts **connection toolbox**, which you can find in the **library** under **standard** ▶ **PythonParts** ▶ **supporting structure**, is opened.

Set the parameters of each element in the two tabs of the **connection toolbox** palette, click the **confirm** button in the **positioning** area at **create element** and place the element at the desired position.

New Object Structural Framing Brace

With the new  **structural brace** function in the **structural framing objects** task area, you can now create braces in addition to columns and beams. The scope of the brace parameters corresponds to that of the beam and it is also generated in the same way as the beam.

Mark Numbers for Structural Framing Object

In the palettes of structural framing objects, the **positioning** section has been added to the **attributes** tab. Here you define the expression for the **schema** as well as the numerical value for the **start number**. These specifications are then used to generate the **position number** with the new  **positioning** function in the **structural framing objects** task area. To do this, after selecting the function, select all the structural framing objects for which you want to generate a mark number and right-click on the workspace to confirm your selection.

To generate the mark numbers, all drawing files that are actively in the background are also taken into account in addition to the actively loaded drawing file. Structural framing objects with identical schema, identical start number and identical parameters and attributes receive the same mark number.

Note: If no expression has been specified for the **schema**, no item number will be generated.

If you modify the structural framing objects, the mark number is displayed for information. You can use the mark number as an attribute of the structural framing object, e.g. for labeling.

Notes on the parameters of positioning

- The random number of characters "\$" in the expression for the **schema** determines with how many digits the number in the mark number is output at least. If, for example, **C4\$\$x** is defined as the schema and **8** as the start number, you get the following mark numbers: C408x; C409x; C410x; ... C499x; C4100x; C4101x; ...
- The expression for the **schema** and the numerical value for the **start number** are stored in the favorites file.
- If you change the default value for the **schema** and/or the **start number** manually, by taking over existing parameters or by loading a favorites file when creating the structural objects, you can use them for all further objects or only for the current object.
- If the **mark number** has already been generated and you change the **schema** and/or the **start number** manually, by taking over existing parameters or by loading a favorites file, the mark numbers already generated are deleted again and must be generated again. A note tells you this.

Control of positioning after modifications

If you have already created mark numbers for the modified structural objects and the modifications have an effect on the correct positioning, a message appears briefly at the bottom right of the screen after you **close** the palette to indicate that objects with incorrect mark numbers exist. You can show this notification permanently by clicking on  **notifications** in the status bar. If you do not want the messages to be displayed automatically for future modifications, activate the **suppress message until project change** option.

If the incorrect positioning is corrected, the icon changes to . This makes it clear that there are no objects with incorrect mark numbers.

Other New Features

Round cross section for structural framing beams

As with structural framing columns, you can now select the  **circle** as the **cross-section shape** for structural framing beams.

Axis Grid Extended and Improved

For the axis grid, you can now enter any designation for the axis grid in the **attributes** area of the **label** tab, which will also be saved in favorites as a property of the grid.

When creating and modifying the axis grid, you can now also create an axis before the first axis. You can do this either intuitively by clicking on the position on the workspace or by entering a negative initial value for the **offsets** in the palette. Furthermore, you can now also delete the first axis.

Intuitive to use and powerful: Terrain model and Road design

The functionality for terrain modeling and road design has been completely redeveloped and includes an intuitive user interface to get started quicker. In addition, the import of point lists, LandXML and REB files as well as the transfer of road alignments from **Bimplus** is supported. To further optimize performance, relevant areas can be cut out of a terrain model. In addition, the number of points in the terrain model can be reduced. Discontinuities in the terrain can be modeled by break lines.

Road design supports parametric modeling of straight lines, transition curves and arcs in site plan and elevation. Slopes are generated independently. Longitudinal and cross sections can be generated and placed automatically according to predefined rules.

The design of utilities placed under or next to roads, such as water, sewage, electricity, Internet, natural gas or district heating, is also supported. Pipes and manholes are generated and displayed in dimensioned and labeled sections.

Introduction to Road Design

You can open a short introduction to the new road planning part of the program as follows:

- On the right side of the title bar, click  **help**, and then click **onboarding street planning**.

Valuable innovations for attribute management

Due to its central importance for BIM projects, attribute management with **Allplan** and **Bimplus** is constantly being further developed. For example, the performance of the interaction between **Allplan** and **Bimplus** has been improved.

The formula editor for attributes now supports the Python programming language as well as a syntax check for detecting incorrect formulas. There is no longer any restriction on the length of attributes.

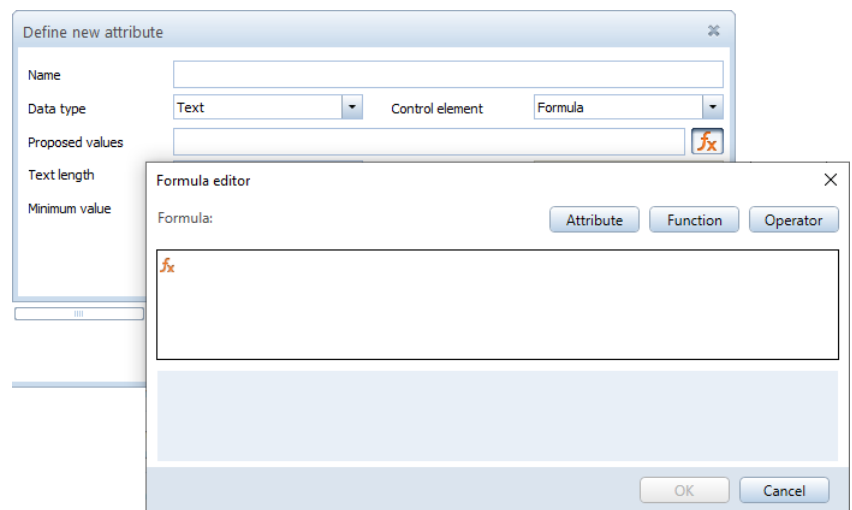
The property palette has also been optimized: Attributes are now grouped into IFC-compliant PSet Commons as well as standard-specific and user-specific attributes. In this case, the associated IFC PSet is automatically activated by assigning the IFC object type. In addition, it is now possible to assign IFC PSets in the building structure.

New Formula Editor

The new **fx** **formula editor** helps you define formulas and formula attributes with a streamlined user interface, better overview, faster attribute selection, and it provides better feedback on syntax inconsistencies.

When defining new attributes, in the **define new attribute** dialog box, select a **data type** for which formulas are possible (all except **date**).

As **control**, select **formula**; then the icon **fx** is displayed with which you open the **formula editor**.



The **formula editor** consists of the following parts:

Attribute

Lists all attributes in alphabetical order. If you click on an attribute, a tooltip with extended attribute name and internal number is displayed; double-click to transfer the attribute with its plain name into the input field.

Function

Lists the available arithmetic functions. If you click on a function, a tool tip with an extended name is displayed; double-click to transfer the function to the box.

Operator

Lists the possible operations. If you click on an operator, a tool tip with an extended name is displayed; double-click to transfer the operator to the box.

Use the keyboard to enter numbers, calculations, and nested parentheses in the box.

Box, formula

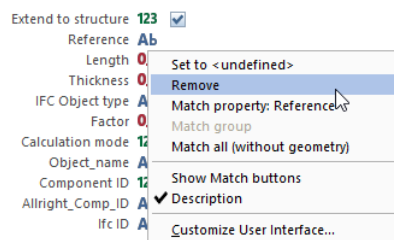
The box displays the selected attributes clear text, functions, and operators. You can also make keyboard entries here; the attributes matching the entered character string are listed and transferred to the input field with a click.

The formula line below displays the attributes selected above with their internal numbers, functions, and operators. Formulas can consist of up to 2048 characters.

If the formula cannot be calculated, the icon  will pop up to the left of the input field. Clicking on the icon puts the cursor in front of the error and you can correct it. If the formula is correct, the icon  will be displayed.

Remove attributes by means of the properties palette

In the shortcut menu in the **properties** palette, you can now remove attributes from one or more components: To do this, select the component(s) and right-click the attribute you want to remove. The **remove** entry, which removes the attribute from the components, is new in the shortcut menu.



Other New Features

Project settings: Path settings for attributes

In previous versions, attribute groups were always stored in the office default, regardless of the path settings for attributes. Now the project path setting applies not only to attributes, but also to attribute groups.

Project settings: Change project for taking over the resources

If you want to change the project from which the project resources are to be taken over in  **new project, open...** in an existing project with the path set to **project**, you subsequently want to change the project from which the project resources are to be transferred, you first need to change from **project** to **office** and then confirm the note and the dialog box.

This resets the project resources to the office default; any changes made will be overwritten.

Then switch back to **project** and use  to select the desired project.

Project Attributes Extended

Attributes for Georeferencing

In the **project properties – attributes** you can now select the following attributes in the **georeferencing** group and assign values to them:

- Coordinate reference system name
- Survey points easting
- Survey points northing
- Survey points height
- Survey points angle of rotation

Export Empty Attributes

Attributes that are assigned to an object but have no entry ("empty" attributes) are *not* exported with  **export quantity data**. Empty attributes are taken into account with  **export attributes**.

Formulas

Formulas can now consist of up to 2048 characters.

Modify Attributes in Building Structure

With **modify attributes** in the shortcut menu of the building structure you can now assign attributes to the structure levels site, building, building and floor or change existing ones or even assign new ones.

Present Compelling Projects

Allplan's integrated visualization capabilities have long been stunning. For version **Allplan 2022**, the graphics engine has been revised so that you can use the features of modern graphics cards even better for better performance and convincing visualizations.

In particular, Vulkan™ technology for NVIDIA® graphics cards offers significant benefits here: NVIDIA® OptiX™ Denoiser uses artificial intelligence to reduce image noise from real-time renderings even faster. This significantly shortens the time to achieve high-quality images. Furthermore, new effects are available using Vulkan™ technology, including volumetric fog, blooming, and lens flare.

Important! These options are only available if your graphics card supports Vulkan™ technology for hardware acceleration (recognizable by the red dot in the title bar of the relevant graphics window). This is usually only the case with newer NVIDIA® graphics cards.

Perspective directly adjustable in window toolbar

You can now switch between parallel and central projection directly from the window toolbar. For this purpose, the new  **perspective** function has been integrated into the window toolbar.

Until now, you had to go via the  **free projection** function (also in the window toolbar) to do this.

Note: As long as the  **perspective** is active, you can select the standard projections, but the model is always displayed in central projection. This is indicated by the absence of the red mark in the projection icon  ( **perspective** function is dominant).

Section display with selected viewing direction

When displaying your construction in an architectural section, you now have the option of retaining the viewing direction set in the window when you select the desired section designation in the **section display** list box in the window toolbar. To do this, click on the desired section designation outside the icon  **representation with the viewing direction of the section**.

If you click the  **display with viewing direction of the section** icon in the list box, **select** the entry and then click the section line or create the section display via the **view** menu, the model data will continue to be displayed with the viewing direction of the selected section.

Optimized RT rendering for NVIDIA graphics cards

For workstations equipped with an NVIDIA® graphics card, the NVIDIA® OptiX™ noise filter offers new visualization possibilities. The noise reduction process makes the most of the NVIDIA® GPU's capabilities, enabling sophisticated, high-resolution renderings to be achieved in the shortest possible time.

The noise filter is available as a downstream post process for the **RT_Render** render process as used in the **RTRender** view type. To do this, select the **NVIDIA OptiX** entry in the **view type** palette for the **RT_Render** render method in the **post process settings** area under **noise reduction filter**.

Important! This option is only available if your workstation is equipped with a graphics card from NVIDIA® that supports Vulkan™ technology (recognizable by the red dot in the title bar of the relevant graphics window).

Other New Features

Tone mapping in view type "animation"

The "tone mapping" for readjusting overexposed areas by reducing the dynamic range (dynamic compression) is now also available in the **animation** view type or all user-defined view types that use the **RT_Render** render method.

Blooming and Lens Effect in Animation View Mode

Provided you are using the **Phong** shading type, the **animation** view type (or the self-defined view types that are based on the **shaded** rendering method) gives you two more new ways to optimize the screen display of your model: **Bloom** and **lens effect**. Both settings are selected from the **view type** palette of the **shaded** render method.

When Blooming is switched on, an additional ring of light is placed around light sources, such as the sun or point-shaped sources of light. The prerequisite is therefore that the **physical sky** has been selected as the background via  **surroundings** (select) or one or more artificial light sources have been set via  **set project light** (select).

When the lens effect is switched on, additional circular light reflections (comparable to the reflection of a light source in a camera lens) are interspersed in the image.

Important! Both effects can only be displayed on your screen if your workstation is equipped with a graphics card that supports Vulkan™ technology (recognizable by the red dot in the title bar of the relevant graphics window). In renderings, both effects are generally *not* taken into account.

Volumetric fog in "animation" view mode

Previously, fog was set via  **options** (**desktop environment** – **animation** tab). This option can now be found in the settings for the  **surroundings** (**visualize** task, **surfaces**, **light** task area). Much more detailed variation options are now available here.

In addition, if the **Phong** shading type is used in the **animation** view type (**shaded** render method), the fog is displayed as *volumetric* fog. This does *not* apply to the other shading types or renderings of the current scenery! For this purpose, a homogeneous fog representation is still used.

Important! Fog can only be displayed volumetrically on your screen if your workstation is equipped with a graphics card that supports Vulkan™ technology (recognizable by the red dot in the title bar of the relevant graphics window).

Light and shadow in the "hidden" and "sketch" view modes

The position of the sun defined via  **surroundings** can be used for the lighting situation in the view types **hidden** and **sketch**. The resulting shadow cast can also be displayed.

Both settings are selected from the **view type** palette of the respective render method **hidden line calculation** or **sketch**.

Physical sky as default background for "animation" and "RTRender" view types

For the predefined view types **animation** and **RTRender**, the default background in the graphics window is now set to **physical sky** (previously **color**).

This setting can be changed via  **surroundings** (**visualize** task, **surfaces**, **light** task area).

Record Movies in Sketch, Wire or Covert Mode

If you want to record movies with  **set camera path** or  **record movie** (both in the **visualize** task, **camera** task area), you can now

also use the rendering methods **sketch**, **wire** and **hidden**, which are familiar from the view types of the same name.

The settings selected for the respective rendering method in the associated *predefined* view type (*not* in a *user-defined* view type) are automatically used as parameters (see "**view type** palette" in Allplan Help).

"Visualize" task now also in "surroundings" role

The **visualize** task is now also included in the **surroundings** role.

Adjust views and sections quickly and easily

With **Allplan 2022**, views and sections have been further optimized. For example, properties can now be easily transferred via the property palette, context menu or wizards. The visibility and representation of objects can be precisely controlled using drawing file and layer filters as well as explicit selection and deselection. The visible area can be quickly and easily adjusted via handles.

Match Properties

When creating views and sections, you can now match the properties of existing views and sections on the workspace or in the wizard by clicking  **match** in the shortcut menu. The reference drawing files of the selected view/section are not adopted.

You can also apply the parameters of an existing view or section when creating or modifying it by clicking  **match properties** in the palette border.

Select Elements for Display

When creating and modifying views and sections, you can now directly select which elements are to be considered for display. To do this, go to the **filter** area, then **entry** and then choose the setting **select**; then you can select the elements you want to be displayed. To confirm, right-click in the workspace.

Please note that depending on the setting you chose for the **entry** option, selecting elements via the button or the  **remove/add elements** function will result in different displays. While the **all** setting for **entry** does not display the elements shown in highlight color, the **select** setting displays only these elements. When you switch from the **all** setting to the **select** setting or vice versa, the selection made using the  **remove/add elements** button or function is reset.

Extended Options for Axis Grid

When defining the position of the axis label for an axis grid that you have created using the  **axis grid** function, you can now additionally define an offset for the starting point of the grid lines in the input options and optionally create a double-sided label.

For views in z-direction, the axes are now completely displayed in x- and y-direction.

Other New Features

Direction Symbol for US Standard Style

If you select the **US standard style**, the selection for the side of the direction symbol has been extended. You can now create the direction symbol in the direction of the viewing direction also on



both sides of the section object.

Section Objects with Handles

You can now intuitively edit the clipping path shown in the section using **direct object modification**. After you have selected it by clicking on it, the usual handles and dialogs are offered on the section object.

Section of Views/ Section

When creating a section from an existing view/section, the entry boxes for **top level** and **bottom level** are now always hidden. This means that you now also have to adjust the height subsequently for sections in the z-direction if necessary.

Better Hatching Style

Georeferenced model data are usually available in rotated form. If sections of this are created in the z-direction and rotated to the horizontal, the hatching is now displayed correctly.

Various enhancements in dimensioning

With **Allplan 2022**, associative dimensioning of angles, radii and arcs is now also possible. In addition, the adaptation of existing dimensions has been optimized. Furthermore, additional texts are possible, and the elevation dimensioning can be rotated.

Curve Dimensioning Updated

The curve dimensioning has been updated and adapted to the linear dimensioning:

- The **associative** option is now also available for curve dimensioning.
- Associative dimensioning of circular and spline walls is now possible with the wall axis switched on.
- The parameter selection has been unified; for this purpose, you will see the dialog box known from straight dimension lines. All parameters can now also be used with curve dimensioning.
- When you create a curve dimension, it is displayed as a preview.
- Curve dimensioning is now processed like linear dimensioning.
- Window height, parapet height and rooms are identified and matched to the dimension line.
- The dimension line parameters are listed in the **properties** palette and can be changed there.
- All curve dimension lines are given the name curve dimensioning (for example, in the **properties** palette, the **objects** palette, or in the **element info**).

Angle Dimensioning Updated

Angle dimensioning has been updated

- The **associative** option is now also available for angle dimensioning.
- The parameter selection has been unified: All parameters known from straight dimension lines can now also be used for angle dimensioning.

Other New Features

Move Dimension Text

In  **move dimension text**, there is now a new **free** button: If this button is pressed, you can place the dimension text at any location. Previously, you had to deactivate the **parallel** and **perpendicular** buttons at the same time to do this.

Additional text for associative dimension lines

You can now enter additional text for existing associative dimension lines.

Select the dimension line. You can see the parameters of the dimension line in the **properties** palette. Select the section for which you want to add an additional text, select the **display additional text** option, and enter the desired text in **additional text**.

Foot as dimension text unit

As a further imperial unit of measure, you can now set the dimensioning to **feet**. The output of the dimension text is done as a decimal number with 4 digits or 2 zeros after the decimal point.

Detect errors more quickly thanks to color coding

As of **Allplan 2022**, objects can not only be quickly filtered via the object palette, but also displayed in color-coded form. This allows component properties to be displayed clearly and checked at a glance. Attributes that have not been assigned or have been assigned incorrectly can thus be recognized quickly.

Objects Palette

Additional Column for Color Coding

To further improve control of the model, you can select a color for all objects in the header of the **objects** palette. You can select from 256 Allplan standard colors using a drop-down field. You can deactivate and reactivate the color with the on/off icon.

Note: This setting affects the **animation** view type only. Make sure that the **use hardware acceleration for all viewports** option is selected in the **hardware acceleration – graphics** area on the **desktop environment – display** page in the  **options**.

You can set the color for the respective group. Individual objects have the same color as the upper or lower group of objects (e.g. openings and/or parapet, window, doorway, etc.).

If the subgroups within an object group (e.g. openings) have different colors (e.g. sill, window, door opening, etc.), an icon for a mixed color

selection appears in the top group.

You can combine color coding and transparency. The operations are analogous to transparency and visibility.

Hide Everything Selected

In the upper part of the **objects** palette, a new button  **hide everything that is selected** has been added.

This allows you to set all elements currently selected on your artboard to invisible.

Filter by Attribute "Style Name"

You can now filter objects (walls) with the **style name** attribute in the **objects -  sorted by attribute** palette. The prerequisite is that the  **attribute to be sorted by** is specified as the first level. This opens the **attribute selection** dialog box, where you can select the **style name** attribute (**architecture general** attribute group).

Reliable basis for cost planning

The determination of traceable quantities is an essential task in the design process. With **Allplan 2022**, not only the quantities contained in the model (e.g. cubic meters of concrete) can be evaluated. It is also possible to output quantities that have not been modeled (e.g. square meters of formwork area). This provides a reliable basis for cost planning.

Reliable Collaboration for OpenBIM Projects

Smooth data exchange is a top priority for ALLPLAN.

Allplan 2022 and **Allplan Bridge** also support the new IFC 4.3 standard. This has been extended specifically for infrastructure structures such as bridges, roads, railroads and waterways. Improved or newly added are the import of data from SketchUp (*.skp) and Wavefront (*.obj), which can transfer data that has both geometric and visual properties. This allows you to import a variety of content objects as well as laser scans (e.g. Zephyr or RealityCapture). In addition, you can now export reinforcement data with sleeves.

New OBJ Format for Importing Models

With  **Import OBJ data** you can now import high-quality models to Allplan that are available as OBJ files. From this, Allplan objects are created. If materials are available, they are also imported and placed directly on the model with the UV coordinates.

You can exchange data with point cloud based surveying systems (e.g. Zephyr, Reality Capture) via OBJ data.

SketchUp Import, OBJ Import

In **Allplan** you can assign only one material to an object. If a SKP or OBJ file contains an object to which several materials or surfaces are assigned, then this object is split into several objects during import depending on the material; sub-objects with the same material are combined again. This will display the object correctly in **Allplan**.

Among others, the following material properties are adopted: Color, texture with color channels, and transparency.

The 3D model is placed on the document with the coordinates defined during creation.

Surfaces and textures are saved in the project under `\design\SKP-Import\model name` or `\design\OBJ-Import\model name`. If identical surfaces are found, you will receive a message. You can replace or keep the existing surfaces.

SketchUp V2021 Importable

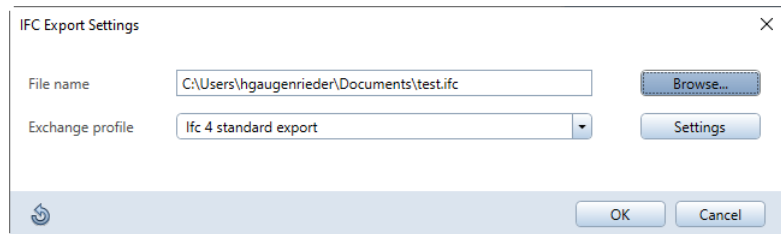
You can now also use SketchUp version 2021 data for imports.

IFC Interface

Improved Interface for IFC Import and IFC Export

The interface of  **Import IFC data** and  **export IFC data** has been improved. The dialogs for file picking and setting options during transfer have been combined and simplified.

If you have already specified the settings or if you use an exchange favorite, you just need to specify the path and filename for the destination – done. If you want to specify export or import in more detail, click **settings**; then the dialog box will expand and you can specify the options in the tabs as before.



Exchange Profiles for IFC Import and IFC Export

Additional exchange profiles have been developed for **IFC export**.

Now you can use the following exchange profiles for export:

- IFC2x3 Coordination View 2.0
- IFC2x3 Standard Export
- IFC4 Reference View
- IFC4 Standard Export
- IFC4.3 Standard Export
- Autodesk Revit
- G&W California IFC2x3
- G&W California IFC4
- Orca IFC2x3
- Orca IFC4

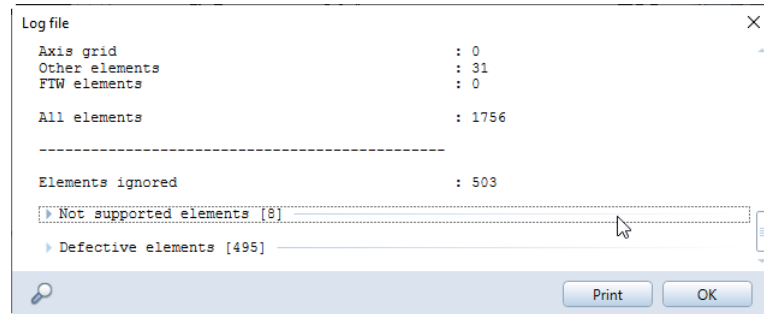
As before, you can use the following exchange profiles for the import:

- Autodesk Revit
- G&W California

Click on the **Information...** button to display further information on the exchange profile.

Enhanced Log

After the data transfer is completed, you will receive a log; at the end of it, ignored elements (e.g. objects without geometry) are listed sorted by type. For a better overview, the individual categories are now displayed collapsed.



IFC Attributes

For the new exchange profile IFC4 reference view and the planned certification, various adjustments and additions were made in Allplan.

IFC attributes for nodes in the building structure

You can now assign attributes to all nodes in the building structure (such as buildings), e.g. for evaluating addresses:

- Department
- Address (street, house number, address suffix)
- Mailbox
- City
- Region
- Zip code
- Country

These attributes will be included when you export IFC data.

3D objects with IFC object type by default

Certain 3D objects are now created with the **IFC object type** attribute by default and are assigned an **undefined** attribute value.

- 3D Line
- Cuboid (pyramid), sphere, cylinder
- User-defined architectural element, converted user-defined architectural element

Further IFC object types

For the new exchange profile IFC4 reference view and the planned certification, various adjustments and additions were made in Allplan.

Length of entire wall

According to IFC documentation, the length of an IfcWall type wall (polygonal or varying thickness) should be the entire nominal length of the wall along the center line of the wall, even if it differs from the wall path. The following changes result for the IFC export of a complete wall:

- The **length @220@** of a total wall in **Allplan** is now calculated in accordance with IFC.
- The **thickness @221@** of a total wall is now calculated as the sum of the thickness of the individual layers.
- For **length_absolute @198@** and **thickness_absolute @199@** of a total wall, the maximum extension in x-y direction in axis direction is now calculated.

IfcMaterialLayerSet

The 'IfcMaterialLayerSet' can now be assigned to parts and directly given a name; now the 'reference' attribute is used for it. The way of assigning the 'codetext' attribute to a part or the first layer of a part is no longer necessary.

IfcType for window and door SmartParts

You can now assign the **IfcType** in the **settings** tab when creating  **window SmartParts** as well as  **door or gate SmartParts**. If you select the **default** option, the detected IfcType will be automatically

selected and displayed. The **user-defined** option allows you to select and assign the desired IfcType.

Geometry attributes of columns

To comply with the IFC standard, the geometry attributes of columns are renamed during IFC export:

- Height (@222@) becomes length
- Length (@220@) becomes width
- Thickness (@221@) becomes depth

Data Exchange with CAFM Programs (e.g. waveware) via DWG

Since **Allplan 2021**, a room stamp can be used to place the base point within a room. Rooms (incl. stamps) are recognized in CAFM programs like **waveware**.

For data exchange with **waveware**, you can now use the new labeling screen **room stamp FM**, which evaluates the designation, function and floor space (without unit of measurement m2) of the room. You can find **room stamp FM** under **select label style – standard** directory – file **1 room stamp**.

Note: In order for the base point to be placed via the room stamp within the room, please first adjust the following registry key:

Select the path

Computer\HKEY_CURRENT_USER\Software\Nemetschek\Allplan\2022.0\Settings\ODX Data, create the

UseRoomReferencePoint entry if necessary, and enter a *nonzero* value for it. Details can be found in the Allplan help under "Transfer rooms to CAFM program waveware, as DWG".

New role in actionbar: Execution of construction work

BIM and digitization are not limited to the planning process, but also play an increasingly important role in construction. Accordingly, the construction process is already given greater consideration during planning and prefabrication. **Allplan 2022** supports this trend. For planning construction sites in particular, new objects such as cranes and concrete pumps are included.

New: Design of bridges from precast girders

With **Allplan Bridge 2022**, the options for parametric modeling have once again been significantly expanded. A special focus is the design of precast girder bridges. Bloss curves are also supported for the design of railroad bridges.

PythonParts, which have been placeable in Allplan Bridge since version 2021-1, are now displayed in detail in Allplan Bridge, including parametric reinforcement and fixtures. Code-based design and checks can now be carried out according to the American AASHTO LRFD standard in addition to the Eurocode.

For improved usability, undo and redo functions are now possible throughout.

New at ALLPLAN: Solution for design and production of precast elements

The integration of the previously independent Nemetschek brand PRECAST SOFTWARE ENGINEERING has added a solution for the design and manufacturing of precast elements to the ALLPLAN portfolio. As Planbar is already based on the Allplan platform, this opens up new possibilities for customers: Architects, engineers and construction companies can transfer BIM models completely digitally to industrial production, including connection to MES and ERP systems. In addition, precast plants can build on existing BIM models from designers and integrate them directly into the production process.

In a first step, the Allplan licensing system is now also available for Planbar. In further steps, work will be done on merging the two products. Customers are already benefiting from precast-specific developments, such as the new openBIM interface IFC4precast or automatically created element plans with sections, dimension lines and labeling.

Multiple optimizations in Bimplus

The open BIM platform **Bimplus** for interdisciplinary collaboration offers a variety of improvements that, for example, speed up the loading of large models, simplify measurement and optimize document management.

In addition, documents and links can now be attached in the Issue & Slideshow Manager and properties in the Issue Manager can be customized. Finally, export of selected objects in IFC format is now possible.

Single Login for Allplan Webservices

The login has been combined for the Allplan web services: Once you are logged in to one web service, you are automatically logged in to all other web services you are registered with (currently still with the exception of Allplan exchange):

- Allplan Connect
- Allplan Bimplus
- Allplan Share
- Allplan Shop
- Allplan Campus

